



考研竞赛凯哥 第16次周考

考试时间:7月20日上午08:30~10:00

讲解时间:7月21日上午08:30~10:00

注1:本次周考的范围为《数二高数部分所有内容》《线性代数》

注2:周考,每周一次,比较频繁,所以题量减半,考试时间减半,且选填题与解答题的比例随机;

月考,每月一次,完全按照考研的标准进行设置.

注3:无论周考还是月考,目的都是查漏补缺和巩固所学知识,所以题目的难度至少持平或略高于真题的平均难度,大家一定不要因为模考成绩不理想而灰心,考前犯的任何错误都应该珍惜!

1. 已知 $\lim_{x \rightarrow 0} \frac{f(x)\tan x + \arcsin x}{x^3} = 0$, 求 $\lim_{x \rightarrow 0} \left(1 + \frac{x^3}{1+f(x)}\right)^{\frac{1}{\sin x}}$.

解: 本题方法很多, 可以用脱帽法, 这里我们采用添项减项.

$$\begin{aligned} 0 &= \lim_{x \rightarrow 0} \frac{f(x)\tan x + \tan x}{x^3} + \lim_{x \rightarrow 0} \frac{\arcsin x - \tan x}{x^3} = \lim_{x \rightarrow 0} \frac{f(x)+1}{x^2} + \lim_{x \rightarrow 0} \frac{\left(x + \frac{1}{6}x^3\right) - \left(x + \frac{1}{3}x^3\right) + o(x^3)}{x^3} \\ &= \lim_{x \rightarrow 0} \frac{f(x)+1}{x^2} - \frac{1}{6}, \text{ 故 } \lim_{x \rightarrow 0} \frac{f(x)+1}{x^2} = \frac{1}{6}. \\ \text{故 } \lim_{x \rightarrow 0} \left(1 + \frac{x^3}{1+f(x)}\right)^{\frac{1}{\sin x}} &= e^{\lim_{x \rightarrow 0} \frac{1}{\sin x} \ln \left[1 + \frac{x^3}{1+f(x)}\right]} = e^{\lim_{x \rightarrow 0} \frac{1}{x} \cdot \frac{x^3}{1+f(x)}} = e^{\lim_{x \rightarrow 0} \frac{x^2}{1+f(x)}} = e^6. \end{aligned}$$

2. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \ln \frac{n+2k}{3n-2k}$

$$\begin{aligned} \text{解: } I &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \ln \frac{n+2k}{3n-2k} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln \frac{1+\frac{2k}{n}}{3-\frac{2k}{n}} = \int_0^1 \ln \frac{1+2x}{3-2x} dx \stackrel{\text{区间再现}}{=} \int_0^1 \ln \frac{1+2(1-x)}{3-2(1-x)} dx \\ &= \int_0^1 \ln \frac{3-2x}{1+2x} dx = - \int_0^1 \ln \frac{1+2x}{3-2x} dx = -I, \text{ 解得 } I = 0. \end{aligned}$$

3. 设连续函数 $f(x)$ 满足 $\int_0^x t[f(x-t) - e^{t^2}] dt = 0$, 则 $\int_0^1 dx \int_0^x f(x)f(y) dy =$ _____.

$$\text{解: } \int_0^x t[f(x-t) - e^{t^2}] dt = 0 \Rightarrow \int_0^x t f(x-t) dt = \int_0^x t e^{t^2} dt \text{ (这里可以暂时不用求出这个积分).}$$

$$\text{对左边的积分使用区间再现公式, 得 } \int_0^x (x-t)f(t) dt = x \int_0^x f(t) dt - \int_0^x t f(t) dt = \int_0^x t e^{t^2} dt.$$

$$\text{两边对 } x \text{ 求导, 得 } \int_0^x f(t) dt + x f(x) - x f(x) = x e^{x^2}, \text{ 代入二重积分, 得}$$

$$\int_0^1 dx \int_0^x f(x)f(y) dy = \int_0^1 f(x) x e^{x^2} dx = \frac{1}{2} (x e^{x^2})^2 \Big|_0^1 = \frac{e^2}{2}.$$



4. 设 $y = y(x)$ ($x \geq 0$) 是由方程 $y^3 + xy = 8$ 所确定的隐函数, 则 $\int_0^7 y^2(x) dx =$ _____.

解: 填 $\frac{31}{2}$.

$$\begin{aligned} x=0 \text{ 时 } y=2, \quad x=7 \text{ 时 } y=1, \quad \text{并从 } y^3 + xy = 8 \text{ 中反解出 } x = \frac{8-y^3}{y}, \quad \text{故} \\ \int_0^7 y^2(x) dx = \int_0^7 y^2(x) d\left(\frac{8-y^3(x)}{y(x)}\right) \stackrel{\text{分部}}{=} y^2(x) \frac{8-y^3(x)}{y(x)} \Big|_0^7 - \int_0^7 \frac{8-y^3(x)}{y(x)} dy^2(x) \\ = 7 - 2 \int_0^7 [8 - y^3(x)] dy(x) = 7 - 2 \left[8y(x) - \frac{1}{4} y^4(x) \right] \Big|_0^7 = 7 - 2 \left[\left(8 \times 1 - \frac{1}{4} \times 1^4 \right) - \left(8 \times 2 - \frac{1}{4} \times 2^4 \right) \right] = \frac{31}{2} \end{aligned}$$

5. 设 $y = y(x)$ 满足微分方程: $y'' \sin x + 2y' \cos x - 2y \sin x = e^x$.

(1) 用 $y = \frac{u(x)}{\sin x}$ 化简该微分方程; (2) 若 $u\left(\frac{\pi}{2}\right) = \frac{\pi}{2} e^{\frac{\pi}{2}}, u'\left(\frac{\pi}{2}\right) = \frac{1}{2} e^{\frac{\pi}{2}}$, 求 $y(x)$.

解: (1) $y = \frac{u(x)}{\sin x} \Rightarrow u(x) = y \sin x \Rightarrow u'(x) = y' \sin x + y \cos x \Rightarrow u''(x) = y'' \sin x + 2y' \cos x - y \sin x$,

故原方程化为 $u''(x) - u(x) = e^x$.

(2) 齐次方程 $u''(x) - u(x) = 0$ 的通解为 $U(x) = C_1 e^{-x} + C_2 e^x$.

设原方程有特解 $u^*(x) = ax e^x$, 故 $[u^*(x)]'' = a(x+2)e^x$, $[u^*(x)]'' - u^*(x) = 2a e^x$, 故 $a = \frac{1}{2}$.

故原方程的通解为 $u(x) = U(x) + u^*(x) = C_1 e^{-x} + C_2 e^x + \frac{1}{2} x e^x$.

将初值条件 $u\left(\frac{\pi}{2}\right) = \frac{\pi}{2} e^{\frac{\pi}{2}}, u'\left(\frac{\pi}{2}\right) = \frac{1}{2} e^{\frac{\pi}{2}}$ 代入, 解得 $u(x) = \frac{\pi}{4} e^{\pi-x} + \frac{x}{2} e^x$.

故 $y(x) = \frac{u(x)}{\sin x} = \frac{\frac{\pi}{4} e^{\pi-x} + \frac{x}{2} e^x}{\sin x}$.

6. 已知 $x^2 - 2ax + 1 - e^x \geq 0$ ($x < 0$) 恒成立, 则 a 的取值范围是 _____.

解: 填 $\left[-\frac{1}{2}, +\infty\right)$.

分离变量, 原不等式等价于 $a \geq \frac{x^2 + 1 - e^x}{2x}$ ($x < 0$), 故 a 只需大于等于 $\frac{x^2 + 1 - e^x}{2x}$ ($x < 0$) 的最大值.

令 $f(x) = \frac{x^2 + 1 - e^x}{2x}$, $f'(x) = \frac{(2x - e^x)2x - (x^2 + 1 - e^x)2}{(2x)^2} = \frac{x^2 - 1 + (1-x)e^x}{2x^2}$,

令 $g(x) = x^2 - 1 + (1-x)e^x$, $g(0) = 0$, $g'(x) = 2x + (-1)e^x + (1-x)e^x = x(2 - e^x) < 0$ ($x < 0$)

故 $g(x)$ 递减, 故 $g(x) > 0$, 故 $f'(x) > 0$, 故 $f(x)$ 递增.

故 $x < 0$ 时, $f(x) < \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{x^2 + 1 - e^x}{2x} = -\frac{1}{2}$, 故只需 $a \geq -\frac{1}{2}$ 即可.

7. 设 $f(x,y)$ 有二阶连续导数, $g(x,y)=f(e^{xy}, x^2+y^2)$, 且 $\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 0}} \frac{f(x,y)+x+y-1}{\sqrt{(x-1)^2+y^2}}=0$.

证明: $g(x,y)$ 在 $(0,0)$ 取得极值, 并判断是极大值还是极小值, 并求出该极值.

解: 由脱帽法, $f(x,y)=-(x-1)-y+o(\sqrt{(x-1)^2+y^2})$, 且 $f(1,0)=0, f'_1(1,0)=f'_2(1,0)=-1$.

$$g'_x=f'_1 \cdot e^{xy} y+f'_2 \cdot 2x, g'_y=f'_1 \cdot e^{xy} x+f'_2 \cdot 2y, \text{ 故 } g'_x(0,0)=g'_y(0,0)=0.$$

$$g''_{xx}=(f''_{11} \cdot e^{xy} y+f''_{12} \cdot 2x)e^{xy} y+f'_1 e^{xy} y^2+(f''_{21} e^{xy} y+f''_{22} 2x)2x+2f'_2;$$

$$g''_{xy}=(f''_{11} \cdot e^{xy} y+f''_{12} \cdot 2y)e^{xy} y+f'_1 \cdot (e^{xy} xy+e^{xy})+(f''_{21} e^{xy} x+f''_{22} 2y)2x;$$

$$g''_{yy}=(f''_{11} \cdot e^{xy} y+f''_{12} \cdot 2y)e^{xy} x+f'_1 \cdot e^{xy} x^2+(f''_{21} e^{xy} x+f''_{22} 2y)2y+2f'_2;$$

$$A=g''_{xx}(0,0)=2f'_2(1,0)=-2, B=g''_{xy}(0,0)=f'_1(1,0)=-1, C=g''_{yy}(0,0)=2f'_2(1,0)=-2.$$

由于 $AC-B^2=3>0$, 且 $A<0$, 故 $g(x,y)$ 在 $(0,0)$ 处取得极值, 且是极大值 $g(0,0)=f(1,0)=0$.

8. A 为 3 阶矩阵, $\alpha_1, \alpha_2, \alpha_3$ 是线性无关的三维列向量, $\alpha_1+\alpha_2+\alpha_3$ 是 A 的属于特征值 5 的特征向量,

且 $A\alpha_1=\alpha_1+3\alpha_2+2\alpha_3, A\alpha_2=2\alpha_1+\alpha_2+\alpha_3$, 求 $|2A^*|$.

解: $A(\alpha_1+\alpha_2+\alpha_3)=5(\alpha_1+\alpha_2+\alpha_3)$, 再结合题干的另外两个等式, 得 $A\alpha_3=2\alpha_1+\alpha_2+2\alpha_3$.

$$\text{令 } P=(\alpha_1, \alpha_2, \alpha_3), P \text{ 可逆, 且 } AP=(\alpha_1+3\alpha_2+2\alpha_3, 2\alpha_1+\alpha_2+\alpha_3, 2\alpha_1+\alpha_2+2\alpha_3)=P \begin{pmatrix} 1 & 2 & 2 \\ 3 & 1 & 1 \\ 2 & 1 & 2 \end{pmatrix}.$$

$$\text{故 } |A|=\begin{vmatrix} 1 & 2 & 2 \\ 3 & 1 & 1 \\ 2 & 1 & 2 \end{vmatrix}=-5, \text{ 故 } |2A^*|=2^3|A^*|=8 \cdot (-5)^2=200.$$

9. 设二次型 $f(x_1, x_2, \dots, x_n)=\sum_{i=1}^m (a_{i1}x_1+a_{i2}x_2+\dots+a_{in}x_n)^2$, $A=(a_{ij})_{m \times n}$, 则二次型 f 的秩等于 ()

A. m

B. n

C. $r(A)$

D. 无法确定

解: 选 C.

本题考查的其实是“逆用矩阵乘法”的技巧, 课上讲过很多遍了.

$$f(x_1, x_2, \dots, x_n)=(a_{11}x_1+a_{12}x_2+\dots+a_{1n}x_n, a_{21}x_1+a_{22}x_2+\dots+a_{2n}x_n, \dots, a_{m1}x_1+a_{m2}x_2+\dots+a_{mn}x_n) \cdot$$

$$\begin{pmatrix} a_{11}x_1+a_{12}x_2+\dots+a_{1n}x_n \\ a_{21}x_1+a_{22}x_2+\dots+a_{2n}x_n \\ \dots \\ a_{m1}x_1+a_{m2}x_2+\dots+a_{mn}x_n \end{pmatrix}=(x_1, x_2, \dots, x_n)A^T A \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

由于 $A^T A$ 为实对称矩阵, 故 $A^T A$ 是二次型 f 对应的矩阵, 故 $r(f)=r(A^T A)=r(A)$.

10. 三阶实对称矩阵 A 满足 $A^2 = 2A$, $f(x_1, x_2, x_3) = \mathbf{x}^T A \mathbf{x}$ 经正交变换 $\mathbf{x} = Q\mathbf{y}$ 化为 $\lambda y_1^2 + \lambda y_2^2 + \lambda y_3^2 (\lambda \neq 0)$,

$$\text{其中 } Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & a \\ 0 & c & 0 \\ b & 0 & 1 \end{pmatrix} \quad (b > 0, c > 0).$$

(1) 求 a, b, c 的值;

(2) 求一个可逆的线性变换 $\mathbf{x} = P\mathbf{z}$, 将 f 化为规范型.

解: (1) 由于 $A^2 = 2A$, 故 A 的特征值只能在 0 或 2 里取, 又 f 经正交变换后的标准型为 $\lambda y_1^2 + \lambda y_2^2 + \lambda y_3^2 (\lambda \neq 0)$,

$$\text{故显然 } A \text{ 的特征值为 } 0, 2, 2, \text{ 对应的特征向量为 } \xi_1 = \begin{pmatrix} 1 \\ 0 \\ b \end{pmatrix}, \xi_2 = \begin{pmatrix} 0 \\ c \\ 0 \end{pmatrix}, \xi_3 = \begin{pmatrix} a \\ 0 \\ 1 \end{pmatrix}.$$

又由于 Q 是正交矩阵, 而正交矩阵是由两两正交的单位向量构成, 并结合 $b, c > 0$, 得 $c = \sqrt{2}, b = 1$.

再根据 ξ_1, ξ_3 正交, 得 $a + b = 0$, 故 $a = -1$. 综上, $a = -1, b = 1, c = \sqrt{2}$.

$$(2) \text{ 由(1)可得, 由正交变换 } \mathbf{x} = Q\mathbf{y} \text{ 得到的标准型为 } 0y_1^2 + 2y_2^2 + 2y_3^2, \text{ 其中 } Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

$$\text{为化为规范型, 只需再令 } \begin{cases} z_1 = y_1 \\ z_2 = \sqrt{2} y_2 \\ z_3 = \sqrt{2} y_3 \end{cases}, \text{ 即 } \begin{cases} y_1 = z_1 \\ y_2 = \frac{1}{\sqrt{2}} z_2 \\ y_3 = \frac{1}{\sqrt{2}} z_3 \end{cases}, \text{ 也即 } \mathbf{y} = M\mathbf{z}, \text{ 其中 } M = \begin{pmatrix} 1 & & \\ & \frac{1}{\sqrt{2}} & \\ & & \frac{1}{\sqrt{2}} \end{pmatrix},$$

则可将二次型化为规范型 $f = 0z_1^2 + z_2^2 + z_3^2$, 故由 $\mathbf{x} = Q\mathbf{y}$ 且 $\mathbf{y} = M\mathbf{z}$ 可知, $\mathbf{x} = QM\mathbf{z}$, 故 $P = QM$.

$$\text{经计算可得, } P = QM = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & \frac{1}{\sqrt{2}} & \\ & & \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ 1 & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \text{ 即为所求.}$$